

LAST NAME: _____

Practice Quiz 06C — MATH 2055 — Fall 2026

Practice for Quiz 06

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

Revised using GPT-6 Astra in the Codex harness.

1. A student claims that for every real a there is exactly one real x with $ax = a$, and gives this argument:

Divide both sides by a to obtain $x = 1$. Therefore there is exactly one solution.

- (a) Identify the invalid step, and describe all solutions when $a = 0$.

 - (b) Add a hypothesis on a that makes the claim true, and prove the corrected claim. Label the existence and uniqueness parts.